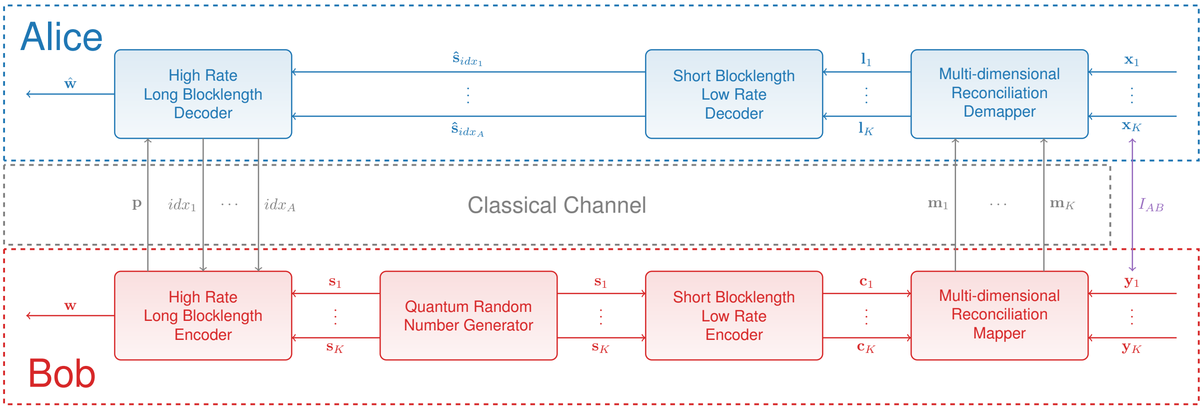

Fig. 1: An overview of short-blocklength reconciliation.

on his side. If the bits do not match, reconciliation fails, and the frame is discarded. If they are the same, Bob confirms this information with Alice and the information bits of the frame $\mathbf{s}$ and $\hat{\mathbf{s}}$ are kept for privacy amplification. Importantly, the revealed bits are discarded when accepting the frame, as these are known to Eve. This has a slight effect on the reconciliation efficiency, as n_{crc} information bits are discarded:

$$\beta = \frac{R \cdot N - n_{crc}}{N I_{AB}}, \tag{1}$$

where I_{AB} is the mutual information between $\mathbf{x}$ and $\mathbf{y}$.

The SKR for multi-dimensional reconciliation can then be given by:

$$\text{SKR} = (1 - \text{FER})(\beta I_{AB} - \chi_{BE} - \Delta_n), \tag{2}$$

where χ_{BE} is the Holevo information, and Δ_n is the finite-size penalty which is dependent on the privacy amplification blocklength $N_{privacy}$[5].

In Fig. 2 we show how n_{crc} influences β depending on N for an $R = \frac{1}{50}$ type-based protograph (TBP) LDPC code[9]. As N decreases, the effect on β_{crc} increases. Even for $n_{crc} = 1$, a reduction in β of 5% occurs when $N = 1000$. Hence, the use of small blocklength error correction codes becomes infeasible. Therefore, we have developed a reconciliation protocol where short blocklength error correction codes can be used, without applying a heavy penalty on β.

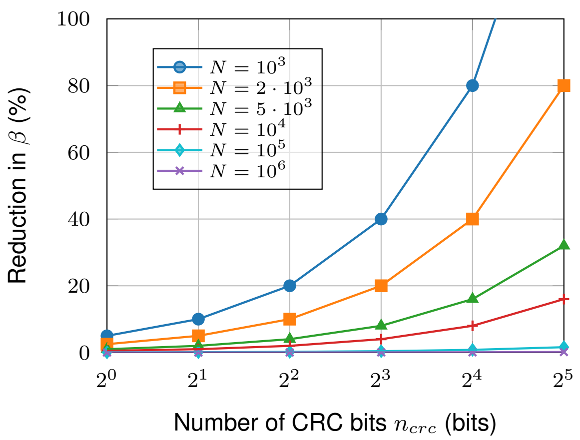

Fig. 2: The reduction in β vs. n_{crc} for different error correction code blocklengths N.

Short-Blocklength Reconciliation

Instead of a CRC, we use a second hard-decision high-rate error correction code, which effectively operates as an outer code, to remove any residual errors between $\hat{\mathbf{s}}$ and $\mathbf{s}$. Let the blocklength of the outer code be N_{out}, which is a multiple of k. We do multi-dimensional reconciliation until $A = N_{out}/k$ frames have been accepted.

The number of reconciliation attempts K for this to happen is on average equal to $\mathbb{E}[K] = \frac{N_{out}}{k \cdot (1 - \text{FER})}$. These bits, $\mathbf{w} = [\mathbf{s}_{idx_1}, \mathbf{s}_{idx_2}, \cdots, \mathbf{s}_{idx_A}]$, where $\mathbf{idx}$ denotes the indices of the accepted frames, are not a codeword of the outer code, hence the syndrome is not equal to $\mathbf{0}$.

Alice sends $\mathbf{idx}$ over the classical channel to Bob. Bob computes the syndrome $\mathbf{p}$ of $\mathbf{w}$ and sends the syndrome under one-time pad encryption to Alice. Alice attempts to recreate $\mathbf{w}$ using the information bits of the accepted frames $\hat{\mathbf{w}} = [\hat{\mathbf{s}}_{idx_1}, \hat{\mathbf{s}}_{idx_2}, \cdots, \hat{\mathbf{s}}_{idx_{N_{out}/N}}]$. As there might be bit flips between $\mathbf{w}$ and $\hat{\mathbf{w}}$, Alice uses $\mathbf{p}$ and the decoder of the outer code to remove residual bit errors. Afterwards, Alice and Bob are left with $\hat{\mathbf{w}}$ and $\mathbf{w}$, which are the same with high probability, and they will use these bits for privacy amplification.

Having to one-time pad $\mathbf{p}$ requires some key material, which has to be deducted from the overall SKR. We can write this as a penalty R_{out} on β:

$$\beta = \frac{R \cdot R_{out}}{I_{AB}}. \tag{3}$$

From our simulations, for all β and N, we observed that the bit error rate (BER) between $\mathbf{w}$ and $\hat{\mathbf{w}}$ is relatively small (BER « 10^{-4}). Hence, the rate of the outer code R_{out} can be very high, $R_{out} > 0.999$, and the penalty to the reconciliation efficiency is negligible.

Simulations

In Fig. 3, we show the FER against the reconciliation efficiency for codes with different blocklengths.